

# Full-Field Rope Solitons

## Sourced Hopf-Link Minima in the Full Rope Field

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### Abstract

Beyond the thin-curve approximation, the Hopf-link soliton can be computed in the full rope field, with the link acting as an extended source. The full-field minimum sits at  $R^* = 0.838$ , consistent with the thin-curve result, showing that the soliton survives the more complete treatment. All numbers in this paper are outputs of the rope\_solver computation in this package (reproducible via benchmarks/reproduce\_results.py), not inserted constants.

#### Scope of this paper

CLAIM: the Hopf-link soliton persists as a minimum in the full rope field at  $R^*=0.838$ , consistent with the thin-curve computation.

NOT CLAIMED: full continuum existence, or particle identification. A computed full-field minimum.

## 1. From Thin Curve to Full Field

The thin-curve treatment models the soliton as a one-dimensional loop. The full-field treatment replaces it with an extended field configuration sourced by the link, testing whether the equilibrium is an artefact of the thin approximation. It is not: the full-field Hopf-link minimum is found at  $R^* = 0.838$ , matching the thin-curve value within tolerance.

#### Status

Full-field Hopf-link minimum  $R^*=0.838$  — Modeled (reproduced, consistent with thin-curve).

Continuum existence — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.